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On-farm water management in saline groundwater area under scarce canal water supply condition in the Northwest India

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ABSTRACT

The study investigates the possibility of enhancing crop water productivity in the parts of Northwest India where groundwater quality is marginal and canal water supply is severely scarce. Soil, Water, Atmosphere and Plant (SWAP) model was calibrated and validated in three farmers' fields with varying canal water availability and groundwater quality in the Kaithal Irrigation Circle of the Bhakra Canal system, Haryana. On the basis of predicted and observed soil water content, pressure heads, salt concentration at 2 week intervals and crop yields, the model was found suitable for use in the region. A few nomographs were prepared to provide a graphical method to predict the effect of different combinations of water quality and depth of water application on crop yield and soil salinity and to help develop some guidelines to the farming community. Water management alternatives at the field level were suggested to increase the yield and to maintain soil salinity below threshold level. The application of frequent irrigation in precisely leveled field would help in achieving 10% higher yield even when saline groundwater of 11 dS/m is used for irrigation.

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1. Introduction

The Northwest India (States of Punjab and Haryana) is one of the most important agricultural regions in the country as it contributes about 52% in nations food production (Abrol, 1999). Rice–wheat cropping sequence is the major food production system in the region, which earlier supported upland crops like maize, cotton and sorghum during kharif/summer (June–October) and wheat during rabi/winter (November–April) seasons. The introduction of rice crop was largely a consequence of waterlogging in canal irrigated areas and development of an intensive network of shallow tubewells. The cultivation of wheat is practiced in about 80% of the area, which even includes marginal quality groundwater areas, mainly due to its low water requirement, tolerance to salinity and consistent market demand resulting in an assured return (Tyagi et al., 2005). Therefore, the rice–wheat cropping sequence in the region evolved due to specific regional climatic and geo-hydrological conditions. Although, the average productivity in the Northwest India is one of the highest in the country (7–8 Mg/ha for the rice–wheat sequence), there are considerable crop yield differences within this region (Ambast et al., 2004a).

In the region, irrigation requirement is essentially met by surface water and groundwater. The canal irrigation system provides water using a rotational system of cycles of fixed duration, frequency and priority level, commonly known as 'warabandi' (Malhotra, 1982). Despite this effective and well-adopted method of water distribution, its supply is unreliable and inadequate to meet the crop water demand. Furthermore,

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the amounts of water received by the farmers do not correspond with the planned allocations (Jacobs et al., 1997). The allocation is very low (1 m^3/s for 5000 ha or 0.2 (l/s)/ha) which could supply about 150–200 mm of water during rice–wheat cropping season. There is a wide variation in canal water availability with respect to location of the irrigated field in the watercourse (Tyagi et al., 2005). The farmers located at the tail end suffer most as they hardly receive any canal water, and the quality of the alternative water source (Ambast et al., 2006), groundwater at depths <40 m, is marginal. In Haryana, nearly two-third and in Punjab, nearly one-fourth of the area has marginal quality groundwater (EC: 2–6 dS/m). In some areas, where groundwater quality is poor (EC > 6 dS/m) and canal water availability is occasional, farmers are mostly confined to growing wheat during the rabi season, though some farmers also grow rainfed pearl millet during the kharif season. However, application of canal water in conjunction with saline groundwater is leading to lower crop yields and increased soil salinity. Khosla et al. (1979) reported that the removal of 80% salts, from 1-m soil depth, accumulated due to saline water application of 6, 9 and 12 dS/m in the rabi season, requires 0.50, 0.75 and 0.90 m of good quality water, respectively. With an average annual rainfall of 625 mm, which occurs mainly during the monsoon (end of June to end of September) in kharif season, and application of 150–200 mm of canal water, keeps the root zone salinity within the acceptable limits. Dhir (1976) and Manchanda and Chawla (1981) reported that monsoon rains leached the salt accumulated in the preceding crop season. Sharma and Rao (1998), who conducted a 7-year field experiment that involved wheat-fallow rotation for first 3 years and wheat-rainfed pearl millet for the last 4 years, confirmed it. Drainage water from subsurface drainage system with salinities of 19 dS/m were used successfully during the rabi season provided the initial electrical conductivity of saturation paste extracts (ECe) of the 0–1.2 m depth interval was less than 4 dS/m, and good quality canal water was applied before wheat was planted. They reported increased soil salinity at the end of rabi season but rain during the ensuing monsoon season reduced this salinity to the original salinity level at the beginning of every rabi season Even if some salts accumulated in the sub-soil after 6–7 years, non-saline canal water could be used to prevent any salt accumulation in the soil profile.

Earlier studies for conjunctive use of canal water with saline water were conducted for well-designed treatments of cyclic and blending mode of irrigations under controlled conditions (Naresh et al., 1993; Sharma et al., 1994; Sharma and Rao, 1998). However, canal water supply is highly unreliable and inadequate in the region (Tyagi et al., 2005) that limits the applicability of recommendations that emerged from earlier studies. Therefore, it is important to consider what actually happens to suggest (i) water management options for improving the yields of wheat grown during the rabi season and (ii) strategies to assess and combat long term impacts of various combinations of saline water applications in different wheat-based crop rotations.

Simulation models provide a means to evaluate various management options (Droogers et al., 2000; Ines et al., 2001). The role of simulation models in understanding the processes in the soil–plant–atmosphere system has increased significantly in recent years. The objectives of this study were to evaluate the impact of conjunctive use of canal and groundwater on wheat yield and soil salinity in high soil salinity areas (initial ECe > 5 dS/m at wheat sowing) and to suggest possible on-farm water management alternatives to increase yields and minimize soil salinity at field level. We used the simulation model, Soil, Water, Atmosphere and Plant commonly referred to as SWAP (Van Dam et al., 1997) to meet some of the objectives of this study.

2. Materials and methodology

2.1. The model

Soil, Water, Atmosphere and Plant (SWAP) has been used in this study due to its versatility. Initially, SWATR a one-dimensional model was developed by Feddes et al. (1978) to describe transient water flow in the heterogeneous soil root system under the influence of groundwater. Subsequently, several researchers improved the model (Belmans et al., 1983; Kabat et al., 1992; Van Dam et al., 1997). It is designed to simulate unsaturated flow, solute transport, heat flow and crop growth in the soil–plant–atmosphere environment at the field scale. The model applies Richard's equation for soil water flow in the soil matrix described as below:

$$C(h)\frac{\partial h}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z} + 1\right)\right] - S(h)$$

where, h is the soil water pressure head (cm), K is the hydraulic conductivity (cm/day), C is the water capacity ($d\theta/dh$) (cm^{-1}), S is the soil water extraction rate by plant roots (cm^3/cm^3)/day), z is the vertical coordinate positive in the upward direction, t is the time (days). SWAP solves the Richard's equation through an implicit finite difference scheme (Van Dam et al., 1997). The model provides options for irrigation scheduling, drainage design, prediction of depth of groundwater table, soil salinity and leaching of nitrogen and pesticides. The model also simulates crop growth using a simple model based on the linear production function of Doorenbos and Kassam (1979) and a detailed crop model WOFOST (Supit et al., 1994). SWAP has been applied successfully in many hydrological studies for a wide range of climatic and agriculture conditions (Kite and Droogers, 2000; Van Dam and Malik, 2003; Qureshi et al., 2004).

2.2. The study area

The study was conducted in the Sirsa branch of Kaithal irrigation circle of the Bhakra Canal system in Haryana, India (Fig. 1.). The area is semi-arid, with an average annual rainfall of 625 mm. About 80% of the annual rainfall is received during the months of June to September. The average annual evaporative demand in the region is about 1600 mm and minimum and maximum temperature varies from 5 to 25 and 12 to 44 °C, respectively. Soil textures range from sandy loams to loams and the organic matter content is generally low.

The availability of canal water is erratic, inadequate and inequitable in the region (Tyagi et al., 2005). Instead of water being available every 14 days for 7 days, it often not available for more than a month and the time interval it is available

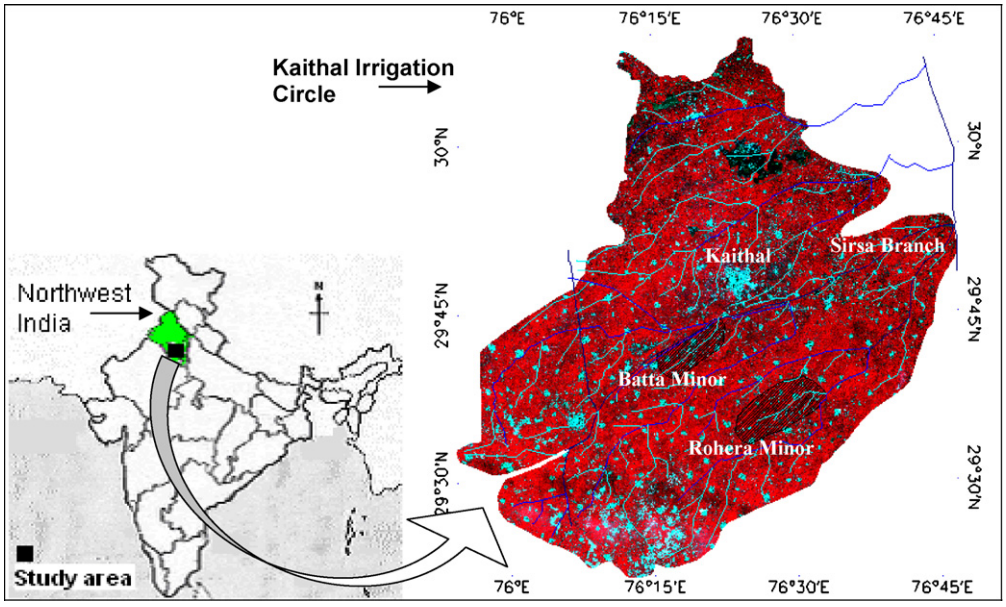


Fig. 1 – Location map and remotely sensed view of the Kaithal irrigation circle, Haryana.

ranges from 3 to 10 days. The major components of soil water balance of rice and wheat crops for estimation of water demand are given in Table 1. The canal water met 32% and 20% of crop water demand of wheat during November to April and rice during June to October, respectively. It was observed that canal water supply seldom satisfied crop water demand during rabi season, whereas it never satisfied the demand in the kharif season (Fig. 2). Further, in the Batta minor (irrigation channel that carries water from the main canal to the watercourse that distributes water among farmer's fields) it was observed that the average canal water supply during the rabi season near the tail of the watercourse was about 56% of that available at the head of the watercourse, whereas it was just 20% in the kharif season. So the quantity and timing of canal water supply limits the scope of water management decisions made by farmers and creates the need for groundwater use.

In the study area, groundwater is used mostly from unconfined aquifers. Most of the farmers use centrifugal pumps driven by 6 hp diesel engines to lift the groundwater. The quality of groundwater is relatively good in the head reaches of the canal system and deteriorates towards tail reaches of watercourse. The analysis of the water samples revealed that the salinity is significantly higher in the tail watercourses (average groundwater salinity >5 dS/m in 60 out of 72 fields) as compared to head and middle watercourses. Fig. 3 shows the water application pattern from different sources: canal and groundwater in different locations near the head, middle and tail of the Batta (BH, BM and BT) and Rohera (RH, RM and RT) minors. Along these minors, groundwater satisfies a major part of the irrigation water requirement.

2.3. *Data collection*

An experiment was conducted in three farmers' fields (Table 2) in the tail watercourse (watercourse number 50820F; design discharge—0.042 m^3/s which provides water for 244 ha) of the Batta minor during rabi season 2003–04. Various input parameters needed for calibration and validation of the SWAP model were collected at 15 days interval. Soil–water pressure heads were measured using tensiometers placed at 25, 50 and 100 cm depths. Soil samples were collected from 25, 50, 75 and 100 cm depths. Gravimetric method was used to determine soil water content, whereas digital EC meter was used for measuring salinity of water and water extracted from saturated soil. Soil texture (sand, silt and clay content) was determined using pipette method, whereas organic matter content was determined using titration method (Walkley and Black, 1934) only once for four different depths.

Irrigation source, time and date of application to each field was recorded during the crop season (Table 2). The average discharge of the tubewells was 10 l/s whereas the watercourse received an average discharge of 30 l/s. The depth of irrigation water was estimated by using discharge and duration of water application. The EC and pH of the irrigation water were determined once in the laboratory for the tubewells in the

Table 1 – Water balance components for wheat and rice crops

Crops	Evapotranspiration (cm)	Percolation (cm)	Rainfall (cm)	Irrigation (cm)
Wheat	33.8	2.0	5.6	30.2
Rice	58.7	85.3	60.6	83.4

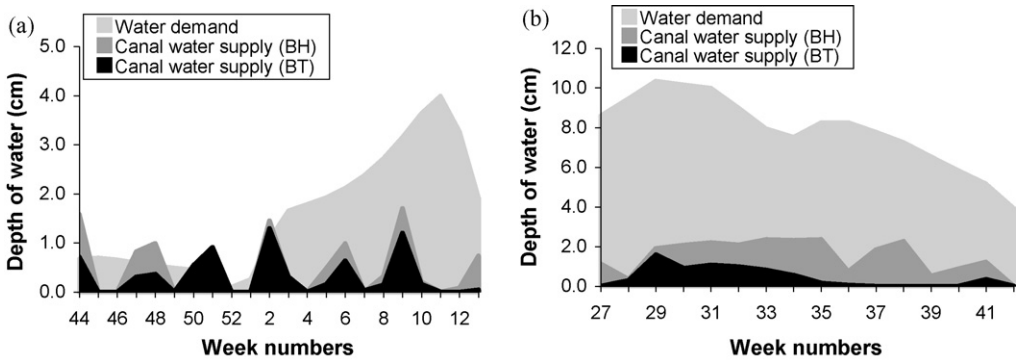


Fig. 2 – Canal water supply and crop water demand during (a) rabi and (b) kharif seasons in the head (BH) and tail (BT) locations of the Batta minor.

three fields. Measurements of crop height and rooting depth were made at 15 day intervals. The rainfall received during the season was measured at the District headquarters at Kaithal, located about 20 km from the fields, whereas other weather data (Fig. 4) were measured at the Meteorology Station of Central Soil Salinity Research Institute (CSSRI), Karnal, located about 60 km from the fields.

2.4. *Calibration and validation of SWAP*

In order to perform simulation studies for various combinations of conjunctive use of canal and ground waters, SWAP was calibrated using observed data in two fields and validated using observed data in the third field. As input parameters for SWAP, top boundary conditions of the soil profile were described by potential evapotranspiration (ET_p), actual rainfall and irrigation on daily basis. The maximum rooting depth for wheat was taken as 110 cm. Root length density distribution was considered to decline linearly with depth. The Boesten model (Boesten and Stroonsnijder, 1986) was used for the reduction of the potential evapotranspiration (ET_p) rate into actual soil evaporation rate (E_a). The calibrated value of Boesten factor was taken as 0.5. The values of pressure heads for regulating root water uptake were taken from Taylor and Ashcroft (1972). These crop parameters are summarized in Table 3. Reference evapotranspiration rate (ET_0) was calculated by the Priestly–Taylor (PT) method (Priestley and Taylor, 1972) as it relies more on radiation rather than on the turbulent momentum, heat and vapour transport mechanisms, and the results are therefore less sensitive to non-representative relative humidity and temperature measurement. Kumar and Bastiaanssen (1993) also suggested the use of PT method instead of the PM for the irrigated areas in India and Pakistan. A rainfall amount of 4.2 cm was considered in all simulations. In the study area, water table was nearly 5 m below ground surface during the entire crop season (Tyagi et al., 2005). Therefore, upward groundwater flux to the root zone was considered zero as the bottom boundary condition.

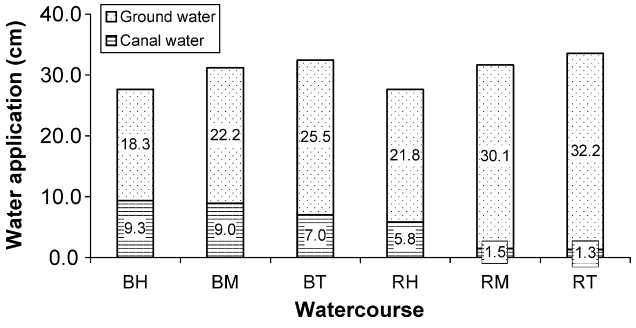


Fig. 3 – Irrigation application by different sources to wheat crop during rabi season (N = 215, Mean = 30.5 cm, S.D. = 5.4 cm) in the head (BH, RH), middle (BM, RM) and tail (BT, RT) locations of the Batta and Rohera minors.

Soil hydraulic properties for a 200 cm deep soil profile was divided into five layers, i.e. 0–17.5, 17.5–35, 35–65, 65–95 and 95–200 cm. Along with average soil texture, the calibrated parameters for all five layers are given in Table 4. The calibrated values of saturated hydraulic conductivity, 20–50 cm/d, were within the range of measured values previously reported for the study area (Ines and Droogers, 2002; Van Dam and Malik, 2003). The salinity parameters in the classical convection–dispersion equation that describe salt transport are dipsersivity (L_{dis}) and the diffusion (D_{dif}). Under field conditions with irrigation, solute spreading due to dispersion is much more pronounced than solute spreading due to diffusion. Therefore, dispersion length L_{dis} was taken as 15 cm while diffusion was taken as 0.48 cm^2/d (Sarwar, 2000).

During calibration, inversion techniques were used to determine soil hydraulic functions. The parameters that were used for calibration and validation of model parameters are: (a) soil water content, (b) pressure heads, (c) EC_e and (d) crop yield. To estimate the crop yield, a simple model based on FAO derived relationships between potential yield, water application and soil salinity was used (Doorenbos and Kassam, 1979). Based on field studies where waters with different salinities were used to irrigate wheat (Sharma and Rao, 1998), the potential yield of wheat crop in the region was taken as 6000 kg ha^{-1} with the slope and threshold values for yield reduction due to time-average soil salinity (ECe for 90 cm) as 8.2 and 3.9 dS/m, respectively. Naresh et al. (1993) reported 9.7, 11.9 and 16.7 dS/m as critical values of ECe for 30 cm at the time of sowing, mid-season and harvest time, respectively that result in yield reductions of 50%. In this study, the simulated yields were compared with measured yields due to several prominent differences namely high initial ECe, actual

Table 2 – The details of the selected fields and water management practices

Description	Field 1	Field 2	Field 3
Location on the tail watercourse	Head	Middle	Tail
Total area (ha)	1.0	0.8	0.8
Crop	Wheat	Wheat	Wheat
Soil salinity (dS/m)	6.8	4.6	7.6
Groundwater salinity (dS/m)	1.7	3.8	9.8
Energy used in pumping	Diesel operated	Diesel operated	Diesel operated
Irrigation dates during the season	20 Oct (Pre-sowing), 13 Dec, 18 Jan, 20 Feb	23 Oct (Pre-sowing), 15 Dec, 20 Jan, 26 Feb	24 Oct (Pre-sowing), 14 Dec, 19 Jan, 15 Feb, 20 Mar
Source/mode of irrigations during the season[a]	TW, CW, CW, TW	TW, TW, CW, CW	TW, TW, CW, CW + TW, TW
Applied water depth (mm)	60, 55, 60, 63	70, 54, 55, 54	70,50, 68, 55, 63
Solute content of applied water[b] (mg/cm^3)	0.65, 0.32, 0.32, 0.65	1.77, 1.77, 0.32, 0.32	6.26, 6.26, 0.32, 3.29, 6.26

[a] Source and mode of irrigation is for the corresponding dates as given in different farmers fields where TW—Tubewell water and CW—Canal water.
[b] Canal water salinity—0.5 dS/m.

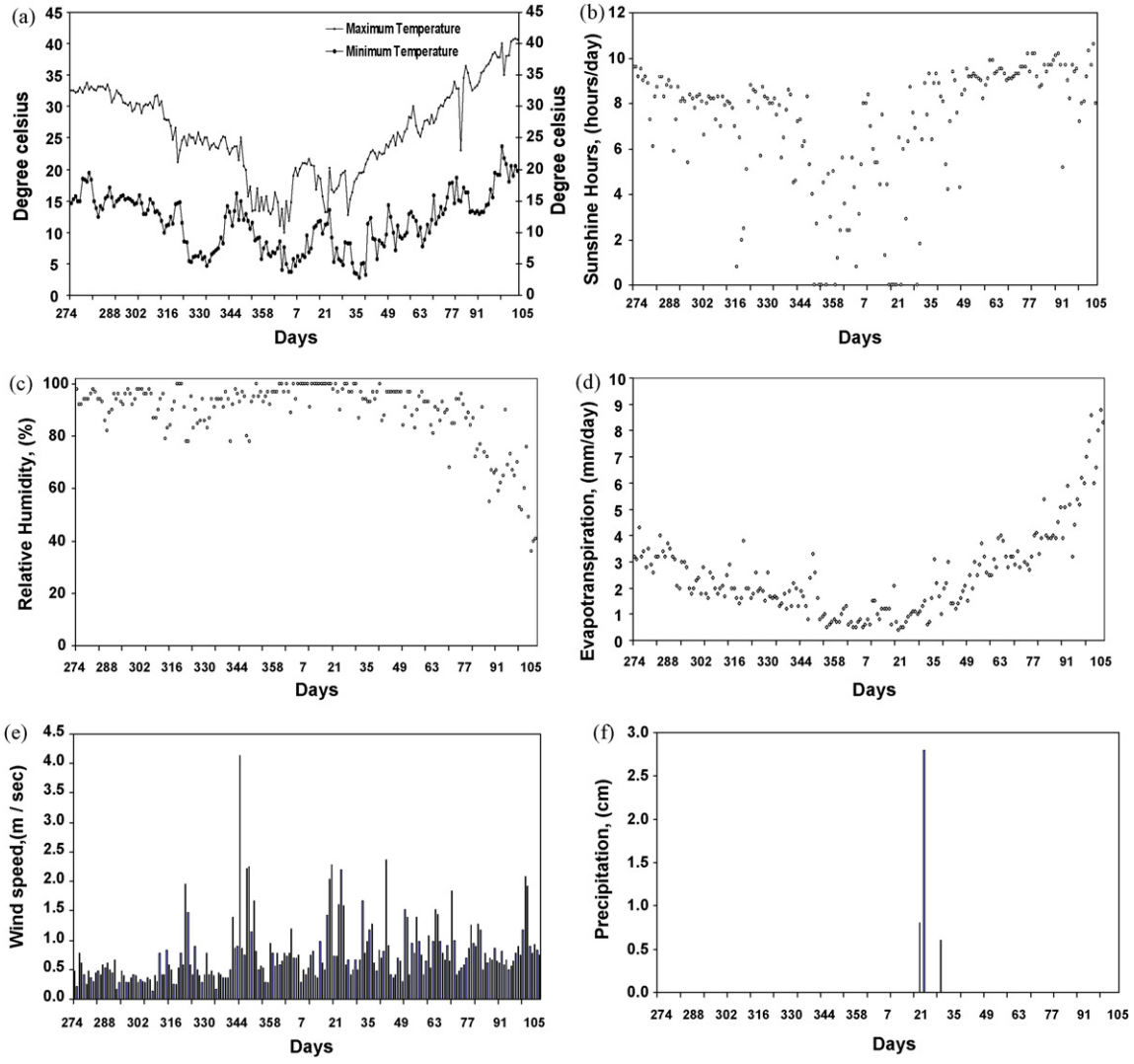


Fig. 4 – (a) Air temperature, (b) sunshine hours, (c) relative humidity, (d) reference evapotranspiration, (e) wind speed and (f) precipitation during 2003–04 in the study area.

Table 3 – Parameters used for calibration in the SWAP model

Sr. No.	Input parameters		Crop—Wheat
1.	Boesten parameter β ($cm^{1/2}$)		0.5
2.	Maximum rooting depth (cm)		110
3.		h_1	−0.1
4.		h_2	−0.1
5.		h_3	−325
6.		h_3'	−500
7.	Limiting pressure heads (cm)	h_4	−8000

on-farm water application (time and depth) and groundwater application (good or saline) at the time of sowing than previously conducted experiments under controlled conditions. The statistical significance of the observed and simulated values was compared using (i) regression coefficient (r^2), (ii) root mean square error (RMSE) and (iii) percentage variation (CV).

Model calibration: Soil water content was determined every 2 weeks at three locations in each field at four depths: 25, 50, 75 and 100 cm. The measured and simulated soil water contents for 25 cm depth in fields 1 and 2 are shown in Fig. 5a. The RMSE for all three depths in field 1 ranged from 0.007 to 0.020 cm^3/cm^3; in field 2 it ranged from 0.015 to 0.034 cm^3/cm^3. The low RMSE values indicate good agreement between measured and simulated soil water content. The CV for observed and simulated values ranged from 0.8 to 4.5% for field 1 and 1.5 to 11.2% in the field 2. The higher CV in field 2 was observed at a depth of 75 and 100 cm (not shown) due to differences in the actual shape parameters (n and λ) and those obtained during calibration. In sum, the model predicted soil water content with sufficient accuracy. The measured and simulated pressure heads at the 25 cm depth for fields 1 and 2 are shown in Fig. 5b. In case of field 1, the simulated pressure heads were in good agreement with observed values for all depths (RMSE ranged 3.7–19 cm). In case of field 2, simulated pressure heads were in good agreement with observed values at 25 cm (RMSE: 18.8) and 50 cm (RMSE: 37.5) depths, but the model over predicted the pressure head at 100 cm depth, which may be due to differences in the actual and obtained shape parameters (n and λ). Some discrepancies were also found at 25 and 50 cm depths where SWAP predicted dryer conditions than the measured values. This was mainly due to the limitation of the tensiometers to read pressure heads below −600 cm. The measured and simulated ECe at 25 cm depth for fields 1 and 2 are shown in Fig. 5c. The model predicted soil salinity with good accuracy for all four depths in both the fields. The RMSE ranged from 0.0 to 0.2 dS/m for field 1 and 0.0 to 0.1 dS/m for field 2. Similarly, CV in simulated and observed ECe was estimated at 0.2–1.4% in field 1 and 0.8–1.5% in the field 2. For field 1, the model predicted the crop yield at 4320 kg/ha as compared to the measured value of 4000 kg/ha (Fig. 5d). For field 2, the predicted yield was 3600 kg/ha as compared to the observed yield of 3500 kg/ha. Thus the model predicted the yield with sufficient accuracy.

Model validation: After calibration of model parameters, these values were kept constant for validation on field 3. The comparison of observed and simulated soil water contents is shown in Fig. 6a. The RMSE was ranged from 0.021 to 0.04 cm^3/cm^3 whereas the CV was estimated from 0.2 to 10.8% for different depths which indicated a good agreement among them. In case of pressure heads (Fig. 6b), SWAP predictions were accurate for shallow depths at 25 and 50 cm: RMSE for observed and simulated values ranged from 16 to 20 cm; the corresponding CV values ranged from 4 to 15%. Some discrepancies between predicted and observed values occurred: SWAP predicted dryer conditions than measured at the 25 and 50 cm depths. This was mainly due to the limitation of the tensiometers to read the pressure heads below −600 cm. At a depth of 100 cm, the model predicted higher than the observed pressure heads. The higher variation observed in the field 3 may be due to change in shape parameters. In case of soil salinity, the RMSE ranged from 0.0 to 0.2 dS/m whereas CV ranged from 0.0 to 0.9%, which indicated a good agreement in simulated and observed values. A comparison of simulated and observed ECe at 25 cm is shown in Fig. 6c. The model predicted crop yield at 2580 kg/ha as compared to the observed yield of 2500 kg/ha (Fig. 6d). Overall, the validation results indicated that the model could predict the field situation with a good accuracy.

3. Results and discussion

After calibration and validation of SWAP model, several scenarios have been generated to obtain a better understanding of the conjunctive use of canal and groundwater as crop/water management practices for wheat production used by the farmers in high salinity areas. The initial soil salinity was taken as 5.5 dS/m whereas salinity of canal water measured as 0.5 dS/m was taken for simulation of different scenarios. The potential yield of wheat crop in the Northwest India is about 6000 kg ha^{-1}, whereas it is about 5500 kg ha^{-1}

Table 4 – Calibrated soil parameters for five different layers

Depth of layer (cm)	0–17.5	17.5–35	35–65	65–95	95–200
Av soil texture (sand:silt:clay)	62.7:18.1:17.8	65.6:14.8:17.6	66.5:17.1:15.5	64.3:17.6:16.1	58.8:18.5:22.6
Residual water content (cm^3/cm^3)	0.0	0.01	0.01	0.0	0.01
Saturated water content (cm^3/cm^3)	0.40	0.365	0.36	0.35	0.37
Sat. hydraulic conductivity (cm/d)	25.02	25.02	25.63	25.45	25.05
Shape parameter (λ)	−1.619	−1.2	−3.0	−2.9	−1.0
Shape parameter (n)	1.22	1.2	1.2	1.1	1.17
Shape parameter (α) (drying) (1/cm)	0.064	0.104	0.1	0.1	0.026
Shape parameter (α) (wetting) (1/cm)	0.047	0.047	0.030	0.026	0.026

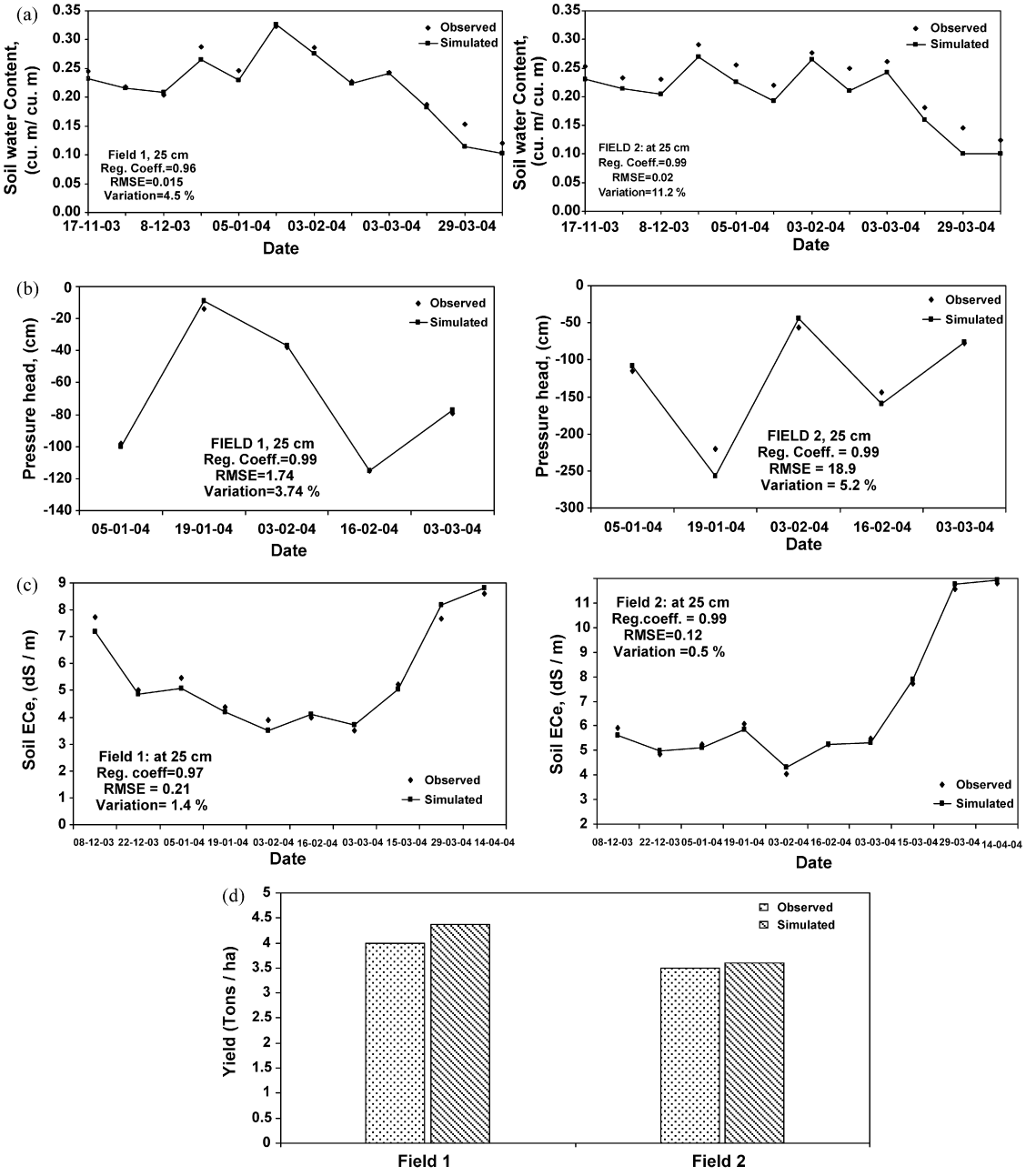


Fig. 5 – Comparison of observed and simulated (a) soil water content (b) pressure heads (c) soil ECe at 25 cm depth and (d) observed and simulated yield.

(relative yield 0.92) in the Kaithal irrigation circle due to marginal quality land and water. In this study, relative yield has been estimated taking potential yield valid for the Northwest India region.

3.1. *Effect of change in water quantity*

The effect of change in quantity of applied irrigation water (10 cm canal water + remaining depth of groundwater; ECgw = 1 dS/m) on relative wheat yield and soil salinity was analyzed (Fig. 7a). It has been observed that relative yield increased from 0.5 to 0.86 with an increase in depth of applied water from 15 to 35 cm, and that a maximum yield could be obtained with 35 cm depth of water application. Further, it has been observed that soil salinity decreased (8.9 to 5.2 dS/m) with increased depth of water application (15 to 40 cm). Applying more than 35 cm of 1 dS/m groundwater might cause a reduction in yield as soil salinity increases thereafter.

3.2. *Effect of change in water quality*

The effect of change in water quality (for 35 cm irrigation depth that includes 10 cm of canal water) on crop yield and soil salinity was simulated (Fig. 7b). It has been observed that

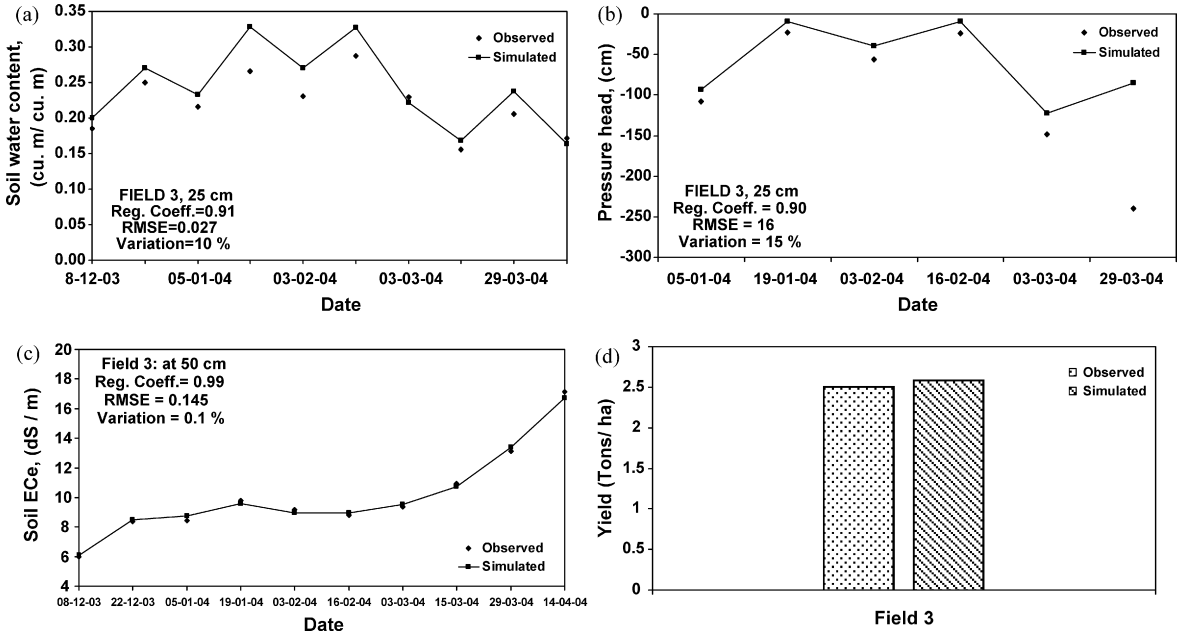


Fig. 6 – Observed and simulated (a) soil water content and (b) pressure heads (c) soil ECe and (d) crop yield in field 3 for SWAP model validation.

relative yield decreased from 0.9 to 0.4 with application of 25 cm of 12 dS/m groundwater and it increased soil salinity from 6 to 19 dS/m. The application of 4 dS/m groundwater slightly increased soil salinity to 6.5 dS/m, whereas the relative yield remained at 0.81. It has also been observed that the application of groundwater with 5 dS/m salinity increased soil salinity considerably (8 dS/m) but the relative wheat yield was reduced only slightly (0.79). Use of groundwater with salinities greater than 5 dS/m decreased crop yield and increased soil salinity considerably. Therefore, groundwater up to 5 dS/m could be applied to wheat crop, as there is not much decrease in the crop yield and increase in the soil salinity.

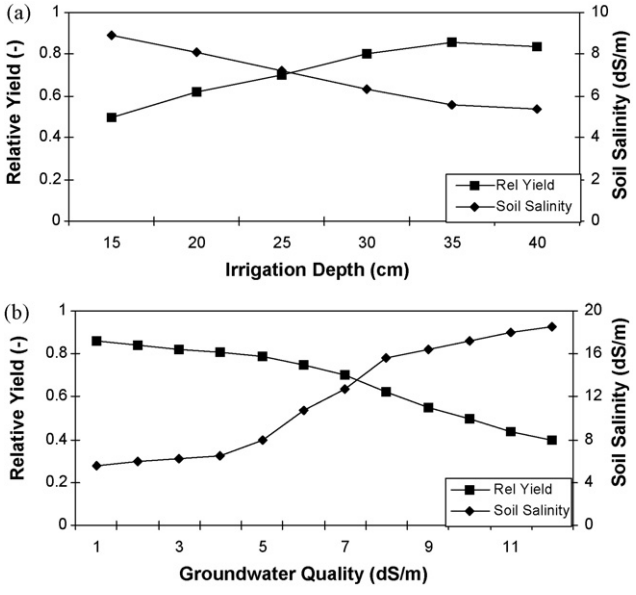


Fig. 7 – Effect of (a) water quantity (ID = Dcw:10 cm + Dgw; ECgw:1dS/m) and (b) quality (ID:35 cm = Dcw:10 cm + Dgw:25 cm) on crop yield and soil salinity (initial soil salinity = 5.5 dS/m).

3.3. *Combined effect of irrigation water quality and quantity*

The effects of different quality and quantity of water application (10 cm canal water + remaining depth of groundwater) on relative yield and soil salinity have been estimated and presented in the form of nomographs (Fig. 8). The development of nomographs may help to suggest some general guidelines for improving crop productivity and minimizing salinity build up with saline water applications in soils with high salinity. Fig. 8a shows the resulting expected relative yield for different depths of water application ranging from 15 to 40 cm and salinity of groundwater ranging from 1 to 12 dS/m. One can see that groundwater quality up to 4 dS/m could be used safely because soil salinity is expected to be less than 5 dS/m and the relative yield is expected to be about 0.8, if 25 cm of groundwater is applied in conjunction with 10 cm of 0.5 dS/m canal water. Further, if 15 cm of 4 dS/m groundwater is applied in conjunction with 10 cm of canal water, the expected relative yield is about 0.6, and remains the same if 20 cm of 4 dS/m groundwater is applied. Furthermore, the relative yield increases significantly when more than 20 cm of groundwater is applied with 10 cm of canal water, but it decreases substantially when more than 25 cm groundwater is applied in conjunction with 10 cm of canal water. This is observed in all the cases of groundwater quality. Less than 30 cm of total water applied results in insufficient leaching whereas salt load in the crop root zone increases significantly

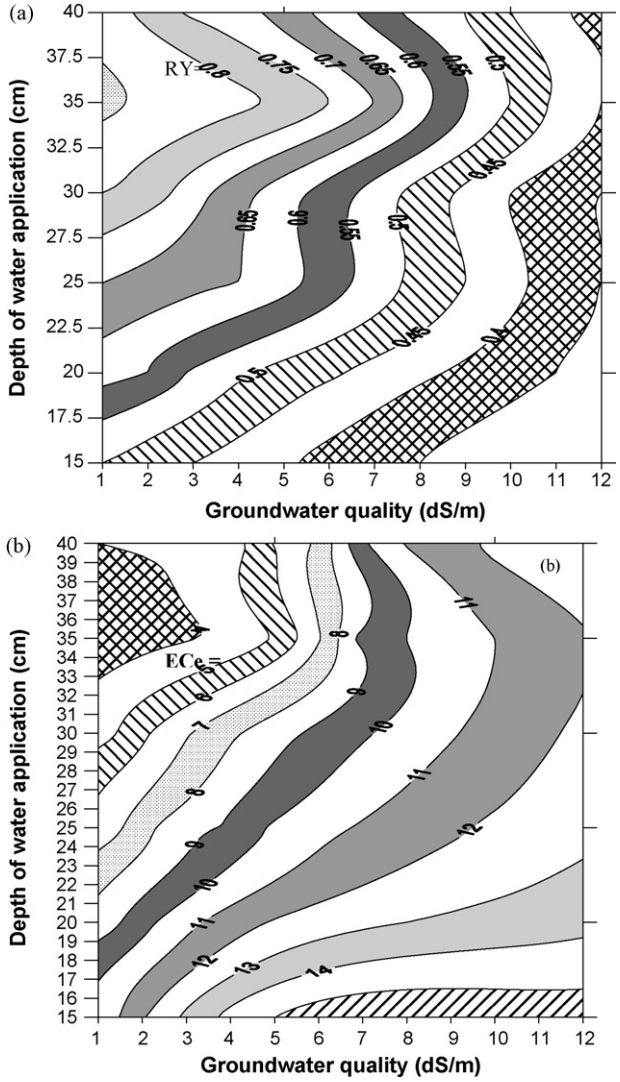


Fig. 8 – (a) Relative wheat yield and (b) soil salinity for different quality of groundwater and quantity of irrigation water application (ID = Dcw:10 cm + Dgw) (initial soil salinity = 5.5 dS/m).

if total water applied is more than 35 cm (Fig. 8b). This clearly indicates that application of 35 cm of total water in place of 30 cm could increase the yield by 20 and 10% in case of groundwater quality of 5 and 10 dS/m, respectively. It will also reduce soil salinity from 10 to 6 dS/m and 12 to 11 dS/m for application of 5 and 10 dS/m groundwater due to better leaching of salts. Irrigation of 10 cm additional water than the normally practiced 25 cm of water application in farmer's fields will increase the pumping cost by 810 INR/ha (pumping cost is taken at 0.81 INR/m^3 for diesel operated pumps; Tyagi et al., 2005) but will give an additional average income of 5000 INR/ha (1 US$ = 45 INR).

Further, crop choices for an area can be made depending upon the quantity and quality of available irrigation water. For example, if threshold soil salinity (ECe) for a crop is 6 dS/m and the salinity of groundwater is 5 dS/m then 36 cm irrigation water would have to be applied as against only 31 cm if salinity of groundwater were 2 dS/m (Fig. 8b). Thus the resultant soil salinity could be controlled by choosing an appropriate combination of quantity of applied water of a given salinity.

3.4. *Effect of canal water availability on crop yield*

Effects of the availability of canal water along with groundwater with different salinities on wheat yield were also simulated. Four different scenarios were developed (i) no canal water supply (Fig. 9a), (ii) 5 cm canal water (Fig. 9b), (iii) 10 cm canal water (discussed in Section 3.3) and (iv) 15 cm canal water. Fig. 9 (a) shows that if canal water availability is limited to 5 cm then 80% of the potential yield can be sustained only when groundwater salinity is 3 dS/m and depth of total applied water is 30 cm, whereas 50% of potential yield could be sustained even with application of saline groundwater up to 10 dS/m. In case of no canal water availability, which is quite common in tail end fields, farmers can sustain 70% of the potential yield with application of groundwater up to 6 dS/m, whereas 50% of the potential yield could be sustained with

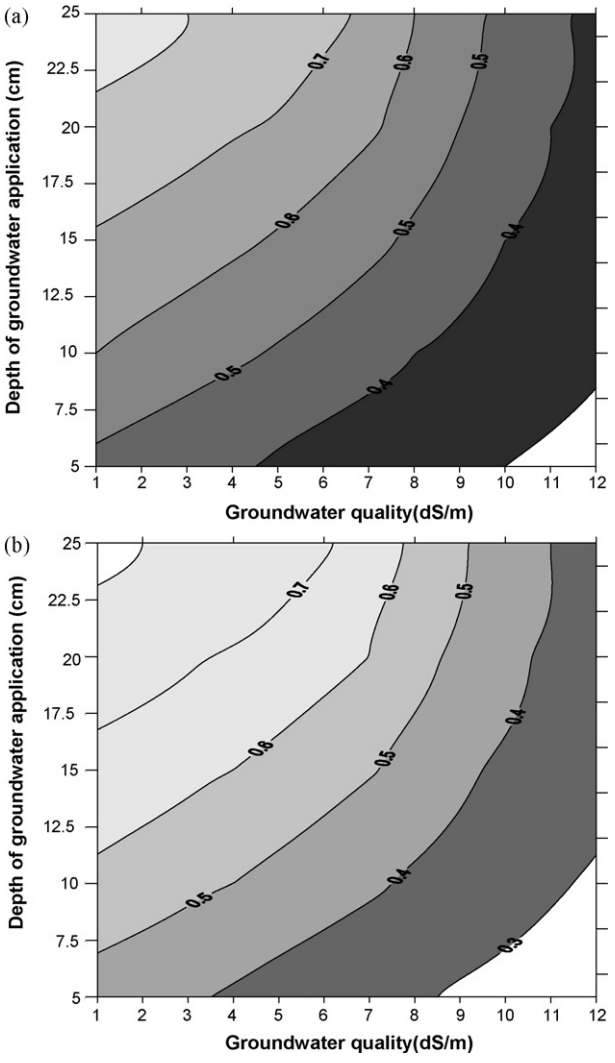


Fig. 9 – Effect of different quality and quantity of groundwater application on relative yield (initial soil salinity = 5.5 dS/m) for canal water application of (a) 5 cm and (b) no supply.

application of 9 dS/m groundwater. Results for different canal water supply indicated that on an average 1 cm increase in canal water supply increased the crop yield by 28 kg/ha. Similarly, an increase in EC of the irrigation water by 1 dS/m causes a reduction in crop yield by 180 kg/ha. Tyagi et al. (2005) reported from the study of field observation data from 216 farmers in the study area that 1 cm increase in canal water increased the crop yield by 24 kg/ha and increase in EC of groundwater by 1 dS/m reduced the crop yield by 153 kg/ha. The simulated values are within the close proximity of the observed value and thus indicating the usefulness of the simulation for farmers field situation.

3.5. *Improving productivity through irrigation scheduling*

Frequent irrigations may help in reducing the salt concentration in the root zone depth and thus improve the crop yield in saline condition. In the study area, one may provide up to six irrigations of 6 cm depth (including pre-sowing irrigation) to the wheat crop in the fields levelled with conventional implements (LI > 1.5 cm; $LI = \left\{\sum_{i=1}^{N}(/L_{di} - L_{ai}/)\right\}/N$ where LI-levelling index representing range of deviations from the desired levels, cm; i—index having value 1 to N, L_d—desired levels at grid points; L_a—achieved levels at grid points and N—number of grid points). In such fields, application of water less than 6 cm is generally observed not feasible with available tubewell discharge of 9–10 l/s. Therefore, irrigation with 6 cm of water per irrigation was considered for estimating the effects of varying number of irrigations of different salinity on wheat yield. It is clear from Fig. 10(a) that relative wheat yield increases more or less in linear fashion with increasing number of irrigations for salinity up to 5 dS/m. However, for irrigation waters greater than 5 dS/m, increases in relative yield with increasing number of irrigations were smaller. By applying six irrigations of good quality water (EC = 1 dS/m) one can get a relative yield of 0.86 that may reduce to 0.4, if salinity of water increases to 11 dS/m.

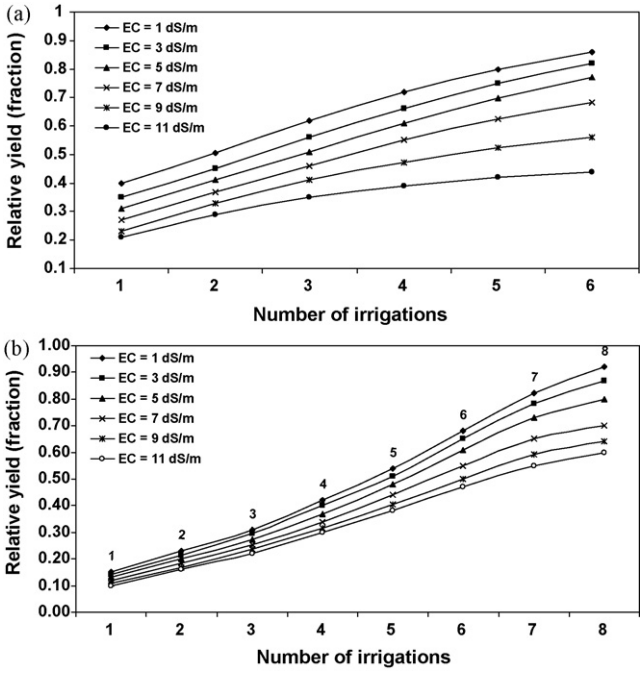


Fig. 10 – Relative wheat yields (initial soil salinity = 5.5 dS/m) due to change in irrigation frequency (a) in conventionally levelled fields (6 cm/irrigation) and (b) in precision levelled fields (4 cm/irrigation).

Laser land leveller helps in achieving precision land leveling (leveling index <1.5 cm) and thus provides the scope of applying shallow depth (4 cm) of irrigation water more frequently (Ambast et al., 2004b). Irrigation scheduling with application of 4 cm water was considered as an improvement option and its effect on crop yield was simulated. It has been observed that relative yield increased from 0.86 to 0.92 with eight irrigations (4 cm) of good quality water (EC < 1 dS/m). After eight irrigations no significant increase in the yield has been observed. In case of 11 dS/m water, the relative yield increased significantly higher (from 0.45 to 0.60) due to frequent irrigation as compared to good quality water. The increase in relative yield has been seen as a result of frequent irrigations that reduced the salinity build up in the root zone depth before it reaches to threshold level. Even the minimum increase in yield (360 kg/ha) alone ensures the net return of 2400 INR/ha (apart from saving money due to reduced pumping time), which offsets the cost of laser land levelling (2500 INR/ha) in the first year itself, whereas the benefit of laser land leveling remains effective for next 2–3 years (1 US$ = 45 INR).

4. Conclusions

In the arid and semi-arid regions, where canal water availability is scarce and groundwater quality is marginally saline, conjunctive use of waters is quite common. Lack of understanding of interaction of different combinations of water quantity and quality on crop yield and soil salinity prohibits the scope for suggesting possible options of improving land and water productivity. In this paper, SWAP model has been calibrated and validated in the Kaithal irrigation circle, Haryana in the Northwest India. It has been observed that model simulates, with sufficient accuracy, the observed field values on soil moisture content, pressure head, soil salinity and crop yield and thus could be used for developing different scenarios of conjunctive water use of canal and ground water. Initially an attempt has been made to analyze the impact of various combinations of water quantity and quality on crop yield and soil salinity. Further, a few nomographs depicting effect of different combinations of waters on crop yield and soil salinity have been developed as ready reference for farmers decision making. This would also help in suggesting the crop choices to the farmers depending on their quality and quantity of water availability to them. The effects of water quantity and quality, for limited canal water availability, on crop yield and soil salinity may also be used for development of spatial decision support system. Further, the farmers' decision-making in regard of water application appears to be rational, however, there is still scope of improving crop productivity through better water management. Precision land leveling which helps in providing less water also facilitates the way for frequent irrigation. Such

practice will maintain low salinity regime in the crop root zone depth and thus increases wheat crop yield.

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